

Midterm Exam

Time: Mar/29/26 Sun 18:00 - 20:00
DURATION OF EXAMINATION: 2 hours

Your exam sheet shall include 6 problems. If not, notify the instructors.

1. (18 pt) Solving a Linear System of Equations

Consider the system of linear equations:

$$\begin{cases} x_1 + 2x_2 + x_3 - 3x_4 = 7, \\ x_2 + x_3 - 2x_4 = 2, \\ 2x_1 - x_2 + x_3 + 8x_4 = -4. \end{cases}$$

- (a) (3 pt) Write the coefficient matrix A and the corresponding augmented matrix.
- (b) (6 pt) Compute the row-reduced echelon form (RREF) of the augmented matrix.
- (c) (3 pt) Using the result in (b), write the solution set of the system.
- (d) (4 pt) Determine the rank and nullity (i.e., dimension of null space) of the coefficient matrix A . Justify your answer.
- (e) (2 pt) Find a basis for the null space of A . (no need to prove it is a basis)

Solution:

- (a) The coefficient matrix is

$$A = \begin{bmatrix} 1 & 2 & 1 & -3 \\ 0 & 1 & 1 & -2 \\ 2 & -1 & 1 & 8 \end{bmatrix}.$$

The augmented matrix is

$$[A|b] = \left[\begin{array}{cccc|c} 1 & 2 & 1 & -3 & 7 \\ 0 & 1 & 1 & -2 & 2 \\ 2 & -1 & 1 & 8 & -4 \end{array} \right].$$

(b)

$$\begin{aligned}
& \left[\begin{array}{cccc|c} 1 & 2 & 1 & -3 & 7 \\ 0 & 1 & 1 & -2 & 2 \\ 2 & -1 & 1 & 8 & -4 \end{array} \right] \xrightarrow{R_3 \leftarrow R_3 - 2R_1} \left[\begin{array}{cccc|c} 1 & 2 & 1 & -3 & 7 \\ 0 & 1 & 1 & -2 & 2 \\ 0 & -5 & -1 & 14 & -18 \end{array} \right] \\
& \xrightarrow{R_3 \leftarrow R_3 + 5R_2} \left[\begin{array}{cccc|c} 1 & 2 & 1 & -3 & 7 \\ 0 & 1 & 1 & -2 & 2 \\ 0 & 0 & 4 & 4 & -8 \end{array} \right] \quad (1 \text{ pt}) \\
& \xrightarrow{R_3 \leftarrow \frac{1}{4}R_3} \left[\begin{array}{cccc|c} 1 & 2 & 1 & -3 & 7 \\ 0 & 1 & 1 & -2 & 2 \\ 0 & 0 & 1 & 1 & -2 \end{array} \right] \\
& \xrightarrow{R_1 \leftarrow R_1 - 2R_2} \left[\begin{array}{cccc|c} 1 & 0 & -1 & 1 & 3 \\ 0 & 1 & 1 & -2 & 2 \\ 0 & 0 & 1 & 1 & -2 \end{array} \right] \quad (2 \text{ pt}) \\
& \xrightarrow{R_1 \leftarrow R_1 + R_3} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 2 & 1 \\ 0 & 1 & 1 & -2 & 2 \\ 0 & 0 & 1 & 1 & -2 \end{array} \right] \\
& \xrightarrow{R_2 \leftarrow R_2 - R_3} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 2 & 1 \\ 0 & 1 & 0 & -3 & 4 \\ 0 & 0 & 1 & 1 & -2 \end{array} \right] \quad (3 \text{ pt}) \text{ (RREF)}
\end{aligned}$$

(c) From the RREF,

$$x_1 + 2x_4 = 1, \quad x_2 - 3x_4 = 4, \quad x_3 + x_4 = -2.$$

Since column 4 has no pivot, let $x_4 = t$ ($t \in \mathbb{R}$) be a free variable. Then

$$x_1 = 1 - 2t, \quad x_2 = 4 + 3t, \quad x_3 = -2 - t.$$

Hence the solution set is

$$\left\{ \begin{bmatrix} 1 \\ 4 \\ -2 \\ 0 \end{bmatrix} + t \begin{bmatrix} -2 \\ 3 \\ -1 \\ 1 \end{bmatrix} \mid t \in \mathbb{R} \right\}.$$

(d) Column 1, 2 and 3 have pivots, so

$$\text{rank}(A) = 3.$$

Since A has 4 columns, by the rank-nullity theorem,

$$\text{nullity}(A) = 4 - 3 = 1.$$

(e) A basis for the null space of A is

$$\left\{ \begin{bmatrix} -2 \\ 3 \\ -1 \\ 1 \end{bmatrix} \right\}.$$

2. (14 pt) Question Answering

(a) (4 pt) Let

$$S = \{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid x_1 - 2x_2 + 5x_3 = 0\}.$$

Find a basis of S and determine the dimension of S .

(b) (2 pt) Let $A \in \mathbb{R}^{m \times n}$ and $\mathbf{b} \in \mathbb{R}^m$. State a necessary and sufficient condition for the system $A\mathbf{x} = \mathbf{b}$ to have at least one solution. The condition shall involve a certain linear space.

(c) (2 pt) Consider the row operation that takes the absolute value of each entry. For instance, $[-1, 2]$ will be transformed to $[1, 2]$. We do not call this operation an elementary row operation because if performed on a row of an augmented matrix, it will change the solution set of the corresponding linear system. Please provide an example where such an operation changes the solution set.

(d) (2 pt) Let $A \in \mathbb{R}^{m \times n}$. Among $\dim C(A)$, $\dim N(A)$, and $\dim N(A^T)$, which two dimensions sum to m ? Briefly explain your answer.

(e) (2 pt) Write down a 3×3 upper triangular matrix that is not invertible.

(f) (2 pt) To evaluate an object, we often compute a total score from a set of feature ratings and their corresponding importance weights. In essence, we are calculating the _____ of the weight vector and the feature vector. What mathematical concept learned in this course shall be filled into this line?

Solution:

(a) Answer: A basis is

$$\{(2, 1, 0), (-5, 0, 1)\}. \quad (3 \text{ pt})$$

The dimension of S is 2 (1 pt).

Reason: Solve $x_1 - 2x_2 + 5x_3 = 0$:

$$x_1 = 2x_2 - 5x_3.$$

Let $x_2 = s, x_3 = t$, then every vector in S has the form

$$(x_1, x_2, x_3) = (2s - 5t, s, t) = s(2, 1, 0) + t(-5, 0, 1).$$

Thus $S = \text{span}\{(2, 1, 0), (-5, 0, 1)\}$. Since $(2, 1, 0)$ and $(-5, 0, 1)$ are linearly independent, they form a basis of S . Thus the dimension of S is 2.

(b) Answer (2 pt): The system has a solution if and only if $\mathbf{b} \in C(A)$. Reason: The vector $A\mathbf{x}$ is a linear combination of the columns of A . Therefore $A\mathbf{x} = \mathbf{b}$ is solvable exactly when $\mathbf{b}$ lies in the column space of A .

(c) Answer (2 pt): Example: Consider the system

$$-x = 1.$$

This gives $x = -1$.

After taking absolute value of each entry, we obtain

$$x = 1,$$

which gives $x = 1$.

Thus, the solution set changes.

Rubrics: Correct example. (1 pt). Justify that the solution set changed. (1 pt)

(d) Answer (1 pt): $\dim C(A)$ and $\dim N(A^T)$.

Reason (1 pt): Let $\text{rank}(A) = r$. Then $\dim C(A) = r$, $\text{rank}(A^T) = r$. Apply the rank-nullity theorem,

$$\dim N(A^T) = m - \text{rank}(A^T) = m - r.$$

Thus their sum is m .

(e) Answer (2 pt): One example is

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Any upper triangular matrix with at least one "zero" on the diagonal would be fine.

(f) Answer (2 pt): Inner product (dot product).

3. (16 pt) Matrix Product, Independence, and Block Matrix Inverse

(a) (3 pt) Suppose $A = \begin{bmatrix} 2 & -1 & 3 \\ 0 & 4 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 \\ -1 & 0 \\ 3 & -2 \end{bmatrix}$. Compute the matrices AB

and $B^T A^T$.

(b) (5 pt) Find the maximum possible number of linearly independent vectors among the following set:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ -1 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ -1 \end{bmatrix}, \quad \mathbf{v}_4 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_5 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}.$$

(c) (4 pt) Let $A \in \mathbb{R}^{m \times n}$ be partitioned as

$$A = [\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n] = \begin{bmatrix} \mathbf{u}_1^\top \\ \mathbf{u}_2^\top \\ \vdots \\ \mathbf{u}_m^\top \end{bmatrix}.$$

Let $B = AA^\top$. Express B in two forms: one using only a_k , the other using only u_i . Do not use the symbol A in your expressions.

(d) (4 pt) Consider a partitioned block matrix $A \in \mathbb{R}^{(n+k) \times (n+k)}$ given by $A = \begin{bmatrix} B & 0 \\ 0 & E \end{bmatrix}$ where $B \in \mathbb{R}^{n \times n}$ and $E \in \mathbb{R}^{k \times k}$. Suppose B and E are both invertible matrices. Does the inverse of A exist? If so, find its explicit inverse; if not, explain why. (Note: To show a certain matrix G is the inverse of A , you need to verify both the left and right product with A are identity matrices.)

Solution:

(a) $AB = \begin{bmatrix} 12 & -2 \\ -1 & -2 \end{bmatrix}$, and (2 pt) $B^\top A^\top = (AB)^\top = \begin{bmatrix} 12 & -1 \\ -2 & -2 \end{bmatrix}$.

(b) Answer: 3.

Reason: Notice the linear combinations: $\mathbf{v}_3 = \mathbf{v}_1 + \mathbf{v}_2$ and $\mathbf{v}_5 = \mathbf{v}_1 + \mathbf{v}_2 + \mathbf{v}_4 = \mathbf{v}_3 + \mathbf{v}_4$. Thus, vectors $\mathbf{v}_3$ and $\mathbf{v}_5$ can be expressed as linear combinations of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_4$. Now we show that v_1, v_2, v_4 are linearly independent. Suppose

$$av_1 + bv_2 + cv_4 = 0.$$

Then

$$a \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \\ 1 \\ -1 \end{bmatrix} + c \begin{bmatrix} 0 \\ 0 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

Comparing coordinates gives

$$a = 0, \quad b = 0, \quad -a + b + c = 0, \quad -b + 2c = 0.$$

From the first two equations we get $a = 0$ and $b = 0$. Then the third equation gives $c = 0$. Therefore $a = b = c = 0$, so v_1, v_2, v_4 are linearly independent.

Hence the maximum possible number of linearly independent vectors is 3.

(c) Form 1: $B = \sum_{k=1}^n \mathbf{a}_k \mathbf{a}_k^\top$

Form 2: $b_{ij} = \mathbf{u}_i^\top \mathbf{u}_j$ for $1 \leq i, j \leq m$, where b_{ij} is the (i, j) entry of B. Equivalently, $B = (u_i^\top u_j)_{1 \leq i \leq m, 1 \leq j \leq m}$.

(d) Answer: Yes, the inverse exists, and $G = \begin{bmatrix} B^{-1} & 0 \\ 0 & E^{-1} \end{bmatrix}$.

Reason: Let the candidate inverse be

$$G = \begin{bmatrix} X & Y \\ Z & W \end{bmatrix}.$$

An inverse must satisfy $AG = I_{n+k}$, which means

$$\begin{bmatrix} B & 0 \\ 0 & E \end{bmatrix} \begin{bmatrix} X & Y \\ Z & W \end{bmatrix} = \begin{bmatrix} BX & BY \\ EZ & EW \end{bmatrix} = \begin{bmatrix} I_n & 0_{n \times k} \\ 0_{k \times n} & I_k \end{bmatrix}.$$

This yields

$$BX = I_n, \quad EW = I_k, \quad EZ = 0_{k \times n}, \quad BY = 0_{n \times k}.$$

By the invertibility of B and E , we can multiply B^{-1} on the left of $BX = I_n$ and $BY = 0_{n \times k}$, and multiply E^{-1} on the left of $EW = I_k$ and $EZ = 0_{k \times n}$, thus

$$X = B^{-1}, \quad Y = 0_{n \times k}, \quad Z = 0_{k \times n}, \quad W = E^{-1}.$$

Thus,

$$G = \begin{bmatrix} B^{-1} & 0 \\ 0 & E^{-1} \end{bmatrix}.$$

Verification: we must explicitly verify $GA = I_{n+k}$:

$$GA = \begin{bmatrix} B^{-1} & 0 \\ 0 & E^{-1} \end{bmatrix} \begin{bmatrix} B & 0 \\ 0 & E \end{bmatrix} = \begin{bmatrix} B^{-1}B + 0 & 0 + 0 \\ 0 + 0 & 0 + E^{-1}E \end{bmatrix} = \begin{bmatrix} I_n & 0_{n \times k} \\ 0_{k \times n} & I_k \end{bmatrix}.$$

Therefore, A is invertible, and its inverse is

$$A^{-1} = \begin{bmatrix} B^{-1} & 0 \\ 0 & E^{-1} \end{bmatrix}.$$

4. (16 pt) True or False

For each question, you do NOT need to justify. (Only writing T or F is enough)

(a) If the columns of an $n \times n$ matrix A span $\mathbb{R}^n$, then A is invertible.

Solution: True.

(b) If a homogeneous linear system has only one solution, then the columns of its coefficient matrix are linearly independent.

Solution: True. Let $A \in \mathbb{R}^{m \times n}$ be the coefficient matrix. Since the only solution is the zero vector, we have $\dim(N(A)) = 0$. By the Fundamental Theorem of Linear Algebra, $\text{rank}(A) = n$, which means the columns of A are linearly independent.

(c) If $A \in \mathbb{R}^{n \times n}$ is invertible, then $(A^{-1})^\top = (A^\top)^{-1}$.

Solution: True. Because $(A^{-1})^\top A^\top = (AA^{-1})^\top = I^\top = I$; $A^\top (A^{-1})^\top = (A^{-1}A)^\top = I^\top = I$.

(d) Every spanning set of $\mathbb{R}^n$ must contain exactly n vectors.

Solution: False. Let $V = \{e_1, 2e_1, e_2, e_3, \dots, e_n\}$, then V spans $\mathbb{R}^n$ but V has $n+1$ vectors.

(e) Suppose $A, B \in \mathbb{R}^{n \times n}$. If A and AB are invertible, then B is invertible.

Solution: True. $B = A^{-1}(AB)$ is product of invertible matrices, hence invertible.

(f) $\{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid (x_1 + x_2 + x_3)(x_1 - 2x_2 + x_3) = 0\}$ is a linear space.

Solution: False. Let S be the given set. Take $u = (1, 1, -2)$ and $v = (2, 1, 0)$. Both belong to S but their sum $u + v \notin S$. Thus S is not closed under addition. (Note that S can be expressed as $\{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid x_1 + x_2 + x_3 = 0\} \cup \{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid x_1 - 2x_2 + x_3 = 0\}$. This shows the union of two linear subspaces is not necessarily a linear subspace.)

(g) If $A\mathbf{x} = \mathbf{b}$ has one solution, then it has infinitely many solutions.

Solution: False.

(h) If two rows of a matrix are identical, then the matrix must not have full column rank.

Solution: False. Consider a 3×2 matrix $\begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$. Rows 1 and 2 are identical, yet the two columns are linearly independent.

5. (16 pt) Linear Algebra in Efficient Attention

In modern neural network architectures such as Qwen 3.5, "linear attention" is a common component. It consists of a sequence of matrix multiplications:

$$Q = XW_Q; K = XW_K; V = XW_V;$$

$$Z = QK^\top V$$

where $X \in \mathbb{R}^{n \times d}$, $W_Q, W_K, W_V \in \mathbb{R}^{d \times k}$.

(a) (2 pt) Determine the sizes (rows $\times$ columns) of Q, K, V and the intermediate matrices $A = QK^\top$ and $S = K^\top V$.

(b) (3 pt) Assume $n = 100$, $d = 25$, $k = 6$, calculate the number of multiplications required in computing $A = QK^\top$ and $Z = AV$.

(c) (3 pt) Under the same condition with (b), calculate the number of multiplications required in computing $S = K^\top V$ and $Z = QS$. Based on your numerical results, sum the multiplications from the two steps in part (b) and compare with the sum from part (c). Determine which computational order is more efficient.

(d) (4 pt) Let $\mathbf{x}_i$ denote the i -th row vector of the input matrix X , and $\mathbf{z}_i$ denote the i -th row vector of the output matrix Z . Suppose the 3rd row of X is a linear combination of its first two rows, specifically $\mathbf{x}_3 = 2\mathbf{x}_1 - \mathbf{x}_2$. Prove algebraically that the 3rd row of Z must satisfy the exact same linear relationship: $\mathbf{z}_3 = 2\mathbf{z}_1 - \mathbf{z}_2$.

(e) (4 pt) Suppose $\mathbf{x}_{n+1} \in \mathbb{R}^{d \times 1}$ and $\hat{X} = \begin{bmatrix} X \\ \mathbf{x}_{n+1}^\top \end{bmatrix} \in \mathbb{R}^{(n+1) \times d}$. Let

$$\hat{K} = \hat{X}W_K, \quad \hat{V} = \hat{X}W_V, \quad \hat{S} = \hat{K}^\top \hat{V}$$

Show that $\hat{S} = K^\top V + W_K^\top \mathbf{x}_{n+1} \mathbf{x}_{n+1}^\top W_V$.

Solution:

(a) Q, K, V are $n \times k$ matrices. A is an $n \times n$ matrix, and S is a $k \times k$ matrix.

(b) Computing $A = QK^\top$ requires 60000 multiplications, and computing $Z = AV$ requires 60000 multiplications.

For $A = QK^\top$: each entry A_{ij} of the resulting $n \times n$ matrix A is the dot product of the i -th row of Q (which has k elements) and the j -th column of $K^\top$ (which also has k elements). Computing this single dot product requires k multiplications. Since A contains n^2 entries, the total number of multiplications is $k \times n^2 = n^2 k$. Substituting $n = 100$ and $k = 6$ shows $A = QK^\top$ is 60000 multiplications.

For $Z = AV$: each entry Z_{ij} is the dot product of the i -th row of A (which has n elements) and the j -th column of V (which also has n elements). Computing one such dot product requires n multiplications. Since Z has $n \times k$ entries, the total number of multiplications for this step is $n \times (n \times k) = n^2k$. Substituting the given values $n = 100$ and $k = 6$ again shows $Z = AV$ needs 60000 multiplications.

- (c) Computing $S = K^\top V$ requires 3600 multiplications, and computing $Z = QS$ requires 3600 multiplications. The computational order in part (c) is more efficient.

For $S = K^\top V$: Each entry S_{ij} of S is the dot product of the i -th row of $K^\top$ (which has n elements) and the j -th column of V (which also has n elements). Computing a single such dot product requires n multiplications. Since S contains k^2 entries, the total number of multiplications required to form S is $n \times k^2 = nk^2$. Substituting $n = 100$ and $k = 6$ shows that $S = K^\top V$ needs 3600 multiplications.

For $Z = QS$: Each entry Z_{ij} is the dot product of the i -th row of Q (which has k elements) and the j -th column of S (which also has k elements). Computing one such dot product requires k multiplications. Since Z has $n \times k$ entries, the total number of multiplications for this step is $k \times (n \times k) = nk^2$. Substituting $n = 100$ and $k = 6$ shows that $Z = QS$ needs 3600 multiplications.

Sum comparison: The total multiplications required for part (b) is $60,000 + 60,000 = 120000$. The total multiplications for part (c) is $3600 + 3600 = 7200$.

Conclusion: Comparing the two sums, $7200 < 120000$. Because it drastically reduces the total number of arithmetic operations, the computational order in part (c) is significantly more efficient.

- (d) Since we can write

$$Z = QK^\top V = (XW_Q)K^\top V = X(W_Q K^\top V).$$

Let $M := W_Q K^\top V \in \mathbb{R}^{d \times k}$, then $Z = XM$. Write this equation row by row

$$\begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{bmatrix} = Z = XM = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} M = \begin{bmatrix} x_1 M \\ x_2 M \\ \vdots \\ x_n M \end{bmatrix}$$

Therefore,

$$z_i = x_i M \quad \text{for each } i.$$

Now use the given relation $x_3 = 2x_1 - x_2$, we obtain

$$z_3 = x_3 M = (2x_1 - x_2)M = 2x_1 M - x_2 M = 2z_1 - z_2.$$

- (e) Since

$$\hat{X} = \begin{bmatrix} X \\ x_{n+1}^T \end{bmatrix},$$

we have

$$\hat{K} = \hat{X}W_K = \begin{bmatrix} XW_K \\ x_{n+1}^T W_K \end{bmatrix} = \begin{bmatrix} K \\ x_{n+1}^T W_K \end{bmatrix}, \quad \hat{V} = \hat{X}W_V = \begin{bmatrix} XW_V \\ x_{n+1}^T W_V \end{bmatrix} = \begin{bmatrix} V \\ x_{n+1}^T W_V \end{bmatrix}. \quad (2 \text{ pt})$$

Therefore,

$$\hat{S} = \hat{K}^\top \hat{V} = \begin{bmatrix} K \\ x_{n+1}^T W_K \end{bmatrix}^\top \begin{bmatrix} V \\ x_{n+1}^T W_V \end{bmatrix}.$$

Now multiply the blocks,

$$\hat{S} = K^\top V + (x_{n+1}^T W_K)^\top (x_{n+1}^T W_V). \quad (1 \text{ pt})$$

Since

$$(x_{n+1}^T W_K)^\top = W_K^\top x_{n+1},$$

it follows that

$$\hat{S} = K^\top V + W_K^\top x_{n+1} x_{n+1}^\top W_V. \quad (1 \text{ pt})$$

6. (20 pt + 5 bonus pt) Rank, Independence

A matrix $A \in \mathbb{R}^{n \times n}$ satisfying

$$A^2 = A$$

is called an idempotent matrix. Let A be such a matrix, and let I_n be the $n \times n$ identity matrix.

- (a) (3 pt) Prove that $(I_n - A)^2 = I_n - A$
- (b) (3 pt) Prove that an invertible idempotent matrix must be the identity matrix I_n .
- (c) (4 pt) Prove that for every vector $x \in C(A)$, we have $Ax = x$
- (d) (5 pt) Suppose $v \in C(A)$ and $w \in N(A)$ are both non-zero vectors. Prove that the set $\{v, w\}$ is linearly independent.
- (e) (5 pt) Prove that $\text{rank}(A) + \text{rank}(I_n - A) = n$.
(Hint: Use the Rank-Nullity Theorem.)
- (f) (Bonus 5 pt) Let $A, B \in \mathbb{R}^{n \times n}$ be two idempotent matrices (meaning $A^2 = A$, $B^2 = B$). Further, assume that $C(A) = C(B)$ and $N(A) = N(B)$. Prove that $A = B$.

Solution:

- (a) Compute

$$(I_n - A)^2 = I_n - 2A + A^2 = I_n - 2A + A = I_n - A.$$

- (b) Assume A is invertible. From

$$A^2 = A,$$

left-multiply by A^{-1} :

$$A^{-1}A^2 = A^{-1}A \Rightarrow A = I_n.$$

(c) Solution 1:

Let $x \in C(A)$. Then $x = Ay$ for some y . Hence

$$Ax = A(Ay) = A^2y = Ay = x.$$

Solution 2:

Write $A = [a_1 \ a_2 \ \dots \ a_n]$ where each $a_i \in \mathbb{R}^n$ is the i -th column vector. Then

$$A = A^2 = A[a_1 \ a_2 \ \dots \ a_n] = [Aa_1 \ Aa_2 \ \dots \ Aa_n].$$

Hence $a_i = Aa_i$ for every $i = 1, \dots, n$. Now take any $x \in C(A)$. There exist scalars $u_1, \dots, u_n$ such that $x = \sum_{i=1}^n u_i a_i$. Then

$$Ax = A\left(\sum_{i=1}^n u_i a_i\right) = \sum_{i=1}^n u_i (Aa_i) = \sum_{i=1}^n u_i a_i = x.$$

(d) Suppose

$$\alpha v + \beta w = 0.$$

Multiply A on the left side:

$$A(\alpha v + \beta w) = \alpha Av + \beta Aw = \alpha v + 0 = 0.$$

Thus $\alpha v = 0 \Rightarrow \alpha = 0$. Substituting back gives

$$\beta w = 0 \Rightarrow \beta = 0.$$

Hence $\{v, w\}$ is linearly independent.

(e) First show $N(I_n - A) = C(A)$.

If $x \in C(A)$, then $Ax = x$, so

$$(I_n - A)x = x - Ax = 0,$$

hence $x \in N(I_n - A)$. This shows $C(A) \subseteq N(I_n - A)$.

Conversely, if $x \in N(I_n - A)$, then

$$(I_n - A)x = 0 \Rightarrow x = Ax,$$

so $x \in C(A)$. This shows $N(I_n - A) \subseteq C(A)$. Therefore $N(I_n - A) = C(A)$.

By rank-nullity:

$$\text{rank}(I_n - A) + \dim N(I_n - A) = n.$$

Since $\dim N(I_n - A) = \dim C(A) = \text{rank}(A)$, we get

$$\text{rank}(A) + \text{rank}(I_n - A) = n.$$

(f) Let $x \in \mathbb{R}^n$. Write

$$x = Ax + (x - Ax),$$

where $Ax \in C(A)$. Because $A(x - Ax) = Ax - A^2x = Ax - Ax = 0$, $x - Ax \in N(A)$. Since $C(A) = C(B)$ and $N(A) = N(B)$, we have

$$Ax \in C(B), \quad x - Ax \in N(B).$$

Since B is idempotent and using the conclusion of part (c),

$$B(Ax) = Ax, \quad B(x - Ax) = 0.$$

Therefore

$$Bx = B(Ax + x - Ax) = B(Ax) + B(x - Ax) = Ax.$$

Since this holds for all $x \in \mathbb{R}^n$, we conclude $A = B$.